What makes someone skilled at performing FNAs?

The physician must be genuinely interested in mastering the technique and performing it often enough to gain expertise. As with any other procedure, "practice makes perfect." It is important to have manual dexterity and good "bedside manners" to make the patient feel comfortable. The ultimate measurement of the skill of the aspirator is the rate of unsatisfactory specimens. A 10% rate of unsatisfactory samples (which indicate technical failure) should disqualify the physician from continuing to perform aspirations.

Why are some patients diagnosed incorrectly?

Because as hard as we try, physicians are not infallible. The errors can result from sampling problems (too few needle-sticks from a large lesion), unsatisfactory samples (due to poor technique), or errors of interpretation. The examination of smears is very demanding and tedious. The pathologist may not see the malignant cells, or there may be very few malignant cells (for example in cystic papillary carcinoma). In other cases, the technical preparation of the smears may be poor. In some cases, there is an overlap of cytologic patterns (cellular adenomatoid nodule, follicular neoplasm, and follicular variant of papillary carcinoma).

I am not worried about the procedure, but I am worried about the results (having cancer). What should I do?

Please remember that most of the time the results are "Benign." If the nodule is malignant, the likelihood of total cure may be excellent if diagnosed early. This should be discussed with your endocrinologist.

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Fine Needle Aspiration of the Thyroid: The Patient's Guide

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This information booklet is provided as an educational service to people being tested for possible thyroid cancer. It is not a substitute for a physician's care, and you should consult with your physician about any medical condition.

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Miscellaneous

Who may perform FNAs?

Any physician who is interested in learning how to perform the procedure properly and who has adequate physical facilities (examining table, sink, counter space, slides, slide-holders, etc.).

- **Endocrinologists and internists** have the advantage that they can perform the FNA immediately after detecting the lump or nodule. Thus, the workup may be more expeditious.
- **Surgeons** should have the advantage of manual dexterity. However, the FNA technique is deceptively simple and most surgeons are not proficient at it. Thyroids are particularly difficult to aspirate, in contrast to breast lesions. Also, surgeons do not have as many patients with thyroid lesions as the endocrinologists, so they lack practice.
- **Pathologists** have the advantage that they can examine the smears immediately under the microscope and assess the adequacy of the specimen. Hence, the rate of unsatisfactory specimens is much smaller. However, there are very few pathologists who are willing to perform this procedure. By training, pathologists are more involved with diagnostic issues rather than interacting with patients. We are the physicians "invisible to patients."
- **Radiologists** have the advantage of being able to aspirate non-palpable lesions and deeply located masses in the chest and abdominal cavities (lungs, liver, etc). They use ultrasound, computerized tomography, and other imaging techniques. However, they are not likely to have sufficiently large numbers of thyroid cases to become proficient at aspirating these glands.

How do you know that the cancer has already invaded the lymph nodes?

In ten percent of patients with thyroid carcinoma the thyroid gland may be normal on palpation. There are no thyroid nodules palpable (or detected). Rather, the cancer may present as a lump in the side of the neck, a metastasis to a lymph node in the neck. The aspirate of this lateral neck mass usually yields brownish fluid. Microscopic examination will reveal the malignant cells.

How long should it take for the results to come back?

Usually the pathologist will report results to the referring physician within 24 to 48 hours. Cases diagnosed as malignant are reported very promptly (within 24 hours and by telephone). Then the referring physician has to decide how to convey the results to the patient. Some do it over the telephone, while others prefer to have the patient come to their offices to discuss results in person. Some patients have told me that their endocrinologist has advised them to "make an appointment to see them one week after the FNA." This seems very reasonable to me.

What about a second opinion?

It will vary depending on who requires it. It may be the pathologist, the endocrinologist, the surgeon, or the patient.
(1) The pathologist may not be absolutely certain of the interpretation, or may "suspect cancer" but does not feel able to make a definite diagnosis.
(2) The endocrinologist and/or the surgeon may receive a pathology report that does not appear to correspond to the clinical findings.
(3) For any reason the patient is uncomfortable with the diagnosis, particularly if possible surgery is involved.

Patients must be aware that doctors are not infallible, and physicians are NOT God. No procedure or result is correct 100% of the time.

To a large extent, the patient may obtain a second opinion if the insurance or the patient is willing to pay for it.

Overview

Your physician has just told you that you have a "lump" (or nodule) in your thyroid gland. Or you have felt a lump in your neck and have made an appointment to see your primary physician. If you are a woman from about 20 to 50 years old, you should understand that this is a common finding. Nevertheless, you may be concerned about the possibility of cancer. In fact, most nodules or "lumps" in the thyroid are NOT cancerous.

The challenge for the physician is to determine which nodules are benign and which nodules are cancerous. The following may provide useful information, but they will NOT determine whether the nodule is benign or malignant:

- How long the nodule has been present.
- Whether it is tender or not.
- Whether it feels soft, rubbery, or firm to the touch.
- The blood tests of thyroid function.
- The response to thyroid hormone suppressive therapy.
- Ultrasound findings.
- Thyroid scans and uptake with radioactive elements.

The only test, prior to surgery, that is proven to differentiate a benign from a malignant nodule is the fine needle aspiration (FNA). Hence, your doctor will tell you that you need a fine needle aspiration of the nodule. What follows is some information about this procedure that will help you understand why and how it will be done.

Based on my experience as a pathologist who has been performing FNAs of palpable masses for 26 years (the last 4 years dedicated primarily to FNAs of thyroid nodules), I will try to answer patients' most frequently asked questions. If you do not find an answer to your specific question, or if something is not clear, please do not hesitate to drop me a note: Yolanda C. Oertel, M.D., Director, Fine Needle Aspiration Service, Washington Cancer Institute (C-1219), Washington Hospital Center, 110 Irving Street, NW, Washington DC, 20010-2975. I will attempt to respond to all queries.

I. Before the FNA:

What is a fine needle aspiration?

It is a simple procedure, similar to drawing blood from your arm. The needle used is thinner than the one for drawing blood, and is attached to a syringe in a syringe holder that allows the operator to apply suction easily. Cells from your thyroid lesion will be extracted through this thin needle. If there is fluid in the "lump," we will drain it. These cells will be smeared (spread) on glass slides, stained, and made ready to be examined under the microscope. After examining all the slides, the pathologist will make a cytologic diagnosis and issue a written report.

Do I really need an aspiration?

An FNA is the only non-surgical method of determining whether your thyroid nodule is benign or malignant.

The "tumor is large and has to be removed." In this instance, do I really need an aspiration (FNA)?

Yes, to avoid surprises for the surgeon. On rare occasions, a patient may have a medullary carcinoma or anaplastic carcinoma of the thyroid. Knowing this in advance will help the surgeon plan accordingly. Additional tests will be needed to exclude the possibility of other tumors (pheochromocytoma) prior to proceeding with the thyroid surgery. In rare instances, cancer from other organs (breast, kidney, lung, etc.) may have spread to the thyroid and may appear as if it had originated there.

Should I have an ultrasound prior to an FNA?

Most of the time it is not necessary. If your physician discovers a nodule and he/she (or the pathologist he/she works with) can perform an aspiration immediately, there is no need to wait for an ultrasound. You may have a diagnosis in 24 hours, and if the nodule is malignant, your surgery can be scheduled within a few days.

Can cancer cells spread (escape) as a result of the FNA?

Surgeons and patients frequently express their concern about "spread of the malignant cells along the needle tract." Needle aspiration has been in use at Memorial Sloan Kettering Medical Center in New York City for over 80 years and at Karolinska Hospital in Stockholm for over 50 years; they have yet to report a single case of such spread.

How much will it hurt after the FNA? Will my neck be sore?

The level of pain during and after the procedure is comparable to that experienced when blood is drawn from a vein in the arm. Usually it is less than that because the needle is thinner. Very few patients report using Tylenol (or equivalents). Please try to avoid aspirin.

What are the recommended steps for someone whose results are reported as inconclusive?

The patient should discuss this interpretation with an endocrinologist. If the physician who ordered the FNA is an internist or family practitioner, I suggest that the patient consult an endocrinologist. If the FNA report is inconclusive because the nodule is cellular but not suspicious of malignancy, then the FNA should be repeated. Some endocrinologists believe that a trial of thyroid hormone therapy with subsequent repeat FNA in 6 to 8 months might be appropriate. It might help to have the smears reviewed by another pathologist (second opinion) with a special interest or training in thyroid diseases who might be able to render a more definite diagnosis. If the diagnosis is "follicular neoplasm," then surgery is indicated.

What is the accuracy rate of FNA?

The rate of accurate results depends on the skill and experience of the physician obtaining the samples and interpreting the smears under the microscope. The figures reported in the medical literature vary considerably and may be difficult to interpret accurately.

Could I have cancer that is not detected on FNA?

Yes, because no technique is 100% accurate. If your lump is large and there is a small focus of cancer (next to it or in it), the cancer may not be detected.

What are the benefits of having an FNA?

If the nodule is benign, surgery can be avoided. If the nodule is malignant, surgery can be planned promptly.

What are the risks, complications, or side effects of FNA?

Complications may be immediate or delayed, but all are rare and almost never serious.

What are the immediate complications of FNA?

- The most commonly reported immediate complication is local bleeding. The patient might notice a bruise or swelling in the neck, with variable degrees of tenderness. This usually disappears in one or two weeks. It is extremely rare that this event may require any medical treatment. I am aware of only one case, aspirated by a surgeon, where severe hemorrhage occurred after a larger blood vessel was damaged, and the patient required immediate surgery.
- Vaso-vagal reaction (fainting). Some patients (particularly men) may faint during the procedure. These are the patients who also faint when they have blood drawn.

What are the delayed complications of FNA?

Among the delayed complications are infection at the site of aspiration and damage to the recurrent laryngeal nerve in the neck.

- Infection (suppurative thyroiditis) rarely has been reported. The patient may experience severe pain and swelling in the neck accompanied by fever and difficulty in swallowing. This requires prompt treatment with antibiotics prescribed by a physician.
- The recurrent laryngeal nerve may be hit by the needle (in which case there is immediate sharp pain) or the nerve may be damaged either by bleeding or edema after the aspiration. This delayed injury will cause a temporary change in the voice and some hoarseness.

Do I need an FNA with ultrasound guidance?

This might be necessary if the nodule is not easily palpable (your physician can't feel it with his fingers), or is defined vaguely (your doctor is not quite sure he can detect it). For example: Your lump was discovered by chance. You attended a "health fair" and had a scan of your neck to check your carotid arteries. The scan revealed that although your carotid arteries were fine, there was a nodule in your thyroid.

Is Fine Needle Aspiration (FNA) the same as a Needle Biopsy?

No, it is not. FNA is a much simpler procedure.

What about a Needle Biopsy?

With this procedure a core of tissue is removed from your thyroid gland by a "Tru-Cut" or "Vim-Silverman" needle, with local anesthesia. The tissue then is processed in the pathology laboratory in the same manner as any other surgical biopsy.

Will I be given local anesthesia before the FNA?

We do not use or inject any local anesthetic. Similarly, no local anesthetic is used when drawing blood from your arm. Although the needle we use for FNA is thinner than that used for drawing blood, we apply an ice pack to numb the skin. Thus the needle prick is a minor discomfort.

What does the procedure consist of?

The skin is cleaned with an alcohol swab (no iodine or betadyne is used); then it is dried with a piece of gauze. Drying the skin avoids the "sting" that is felt when the skin is pierced if the alcohol has not evaporated from its surface. Then the needle is inserted.

How long will I spend in the doctor's office for an FNA?

The procedure usually takes about 20 minutes. Each "needle stick" only requires a few seconds. But be prepared to spend 30 to 40 minutes in the doctor's office, because you will need to fill in registration forms, insurance forms, etc., prior to being aspirated.

Should I fast before the FNA?

You do not need to skip a meal before the procedure. There are no restrictions on eating, either before or after the FNA.

Should I stop my medications before the FNA?

No, you do not need to stop any prescription medications. However, it is preferable to stop taking aspirin one or two days before and after the procedure. Please consult with your physician before stopping any medications.

I am on "blood-thinners." Should I stop taking them?

Please consult with the physician who prescribed the anticoagulants. He will advise you on the safety of discontinuing such medication. If this cannot be stopped for a couple of days prior to the FNA, make sure that the physician performing the FNA is aware of this medication, so special precautions may be taken.

Can I wear a "turtle-neck" or a scarf?

When having a thyroid aspiration, you have to uncover your neck. It is easier for the doctor to perform the procedure if your clothing allows easy exposure of your neck. If you wear a turtleneck, it should have a zipper in back or front that will allow the collar to fold down. Better yet, wear a blouse without a collar, or one with buttons in front, or any clothing that will allow easy exposure of your neck. Also, avoid wearing jewelry around your neck.

How can I find out if the physician is skilled (or qualified) at performing FNAs?

Ask an endocrinologist or thyroidologist about this. To whom do they send their patients for FNAs? Or to whom would the doctor (or a member of the doctor's family) go if an aspiration was needed? Also, ask friends or acquaintances who have had the procedure whether they were satisfied with whomever performed it. Usually a "good aspirator" will be associated with a large medical institution.

What is a false negative diagnosis?

This occurs when the cytology report says that your nodule is benign and subsequent events prove that it was a cancer. This varies from one institution to another and has been reported from 1% to 10%. The drawback of a false negative diagnosis is the delay of treatment.

What is a false positive diagnosis?

This occurs when the cytology report says that your nodule is malignant, but after you undergo surgery, no cancer is found. This also varies from one institution to another and has been reported from 1% to 5%. The drawback of a false positive diagnosis is that the operation may not have been necessary.

Should I get a second opinion?

You always have the right to a second opinion. You should do this if you wish and if it will help you feel more comfortable. Remember that you must be willing to pay for it because it might not be covered by your medical insurance. If you have been diagnosed with cancer, a second opinion that confirms the diagnosis may make you feel more comfortable. See page 12 for more information.

Why does an FNA have to be repeated?

The most frequent cause for a repeat FNA is a previous UNSATISFACTORY or "nondiagnostic" aspirate. There are other instances in which the FNA needs to be repeated:

- A nodule that enlarges while you are taking thyroid medication (which was prescribed to shrink the nodule).
- Follow-up of a previous inconclusive report.
- The appearance of a new nodule.
- Finding an enlarged lymph node adjacent to the thyroid gland.
- We recommend another FNA of persistent benign nodules one year later.

When laboratory results do not clarify the case for the clinician or appear to be at odds with the clinical picture, they should be repeated.

What happens next after my FNA is diagnosed as benign?

If your aspirate has been diagnosed as BENIGN, your physician will discuss with you whether you need to be given thyroid medication or not. Most of the time, if your thyroid function tests (your "blood-work") are normal, you will not require any medication. In other instances, you will need thyroid hormone "suppressive therapy," anti-thyroid drugs, or radioactive iodine.

What happens next after my FNA is diagnosed as malignant?

If your aspirate has been diagnosed as MALIGNANT, you will need surgery. The extent of the operation should be discussed with your endocrinologist and surgeon.

What happens next after my FNA is diagnosed as inconclusive?

If your aspirate is INCONCLUSIVE, it may be the result of a scant or limited sample. It is wise to have it repeated by a more experienced physician who may obtain more abundant material. It may be due to lack of experience of the pathologist evaluating the smears. In this case sending the slides for a second opinion might solve the problem. It might occur because the thyroid mass is very cellular. Sometimes there is an overlap in the cytologic appearance of the benign and malignant follicular tumors. In such instances, your physician might consider it appropriate to give you thyroid hormone pills for 6-8 months and then repeat the FNA. If the nodule or lesion is large or you cannot tolerate the uncertainty of not having a definite diagnosis, you might decide to undergo surgery. See page 11 for more information.

What happens next after my FNA is diagnosed as unsatisfactory?

If your aspirate is diagnosed as UNSATISFACTORY or is said to be "non-diagnostic," you need another FNA.

How do I find out if the pathologist is experienced in diagnosing thyroid cancer?

If you have friends or coworkers who have had thyroid cancer, ask them. Go to the library and check publications such as "Who's Who." Ask the librarians, or if you are proficient at literature searches, find out the names of the pathologists who have published scientific articles on thyroid cancer.

II. During the FNA:

How long will the needle be in my neck?

After inserting the needle, if the lesion is solid, the needle will be withdrawn in 2 or 3 seconds If there is fluid in the lesion, it will take longer (maybe 5 to 8 seconds).

Can I breathe while the doctor performs the procedure?

Yes, you can and should breathe while your doctor performs the procedure. However, you should NOT talk or swallow while the needle is in your neck.

How many "needle-sticks" are recommended to get a good sample?

There is considerable clinical judgement involved by the physician performing the FNA. In general, the size of the nodule or mass will determine the number of aspirates. An average of 3 needle-sticks should be done in nodules that are one to two centimeters (cm) in diameter. More aspirates are required for larger masses. If the lesion is cystic (contains fluid) and the first aspirate drains it completely and there is no residual mass palpable, then there is no need for a second needle-stick. If the lesion is firm and solid, then three or more aspirates are desirable to rule out a carcinoma.

III. After the FNA:

Will I need a bandage after the procedure?

No. Most of the time not even a "band-aid" is necessary.

May I drive after undergoing an FNA?

Yes, you may. An FNA is comparable to having blood drawn from your arm.

Will I need a painkiller after the FNA?

Most patients do not need any analgesics after the FNA. Some patients have a very low threshold for pain and take a Tylenol. Aspirin should be avoided.

Will I be able to exercise or "work out" after the procedure?

Yes, you may exercise or work out after the FNA. You may swim, use a treadmill, etc., but try to avoid heavy weight lifting.

Should I plan on taking the day off from work after the FNA?

You may go back to work after the procedure. There are no restrictions.

What happens after the FNA is performed?

An endocrinologist, surgeon, internist, or a pathologist may have performed your aspiration. Regardless of who performed the aspiration, the sample obtained will be sent to a pathology laboratory. A laboratory technician will stain the slides, and a cytopathologist will examine them with a microscope and make a cytologic diagnosis. The cytopathologist then will issue a written report to communicate these findings to the physician who sent the slides or who referred you for FNA.

What is a pathology report?

A pathology (or cytopathology) report is a standardized written document that states what was found on the smears. In other words, it is a cytologic diagnosis -- the results of your aspiration. Your treatment will depend upon it.

What are the possible cytologic results of the FNA?

Usually the results will fall into one of four categories: benign (70-75% of cases), malignant (4-7%), inconclusive (10-15%), and unsatisfactory (1-10%). We will explain the meaning of each of these.

What is a benign diagnosis?

This implies that your nodule or lump is not a malignant tumor. It could be due to inflammation (thyroiditis), accumulated secretion from your gland (colloid nodule), irregular growth of your gland (hyperplastic nodule), or a cyst (fluid-filled nodule).

What is a malignant diagnosis?

This implies that your nodule is cancerous. There are several types of thyroidal cancers. The most common and the easiest to treat successfully is papillary carcinoma.

What is an inconclusive diagnosis?

There is no certainty about the nature of your nodule; it could be either benign or malignant. Please make sure this is not confused with a benign diagnosis. This means that not much progress was made in determining the nature of your lump. Either the FNA has to be repeated or the possibility of surgery should be discussed with your doctor.

What is an unsatisfactory diagnosis?

Whoever performed the FNA was not successful in obtaining enough cells from your thyroid to allow the pathologist to make a diagnosis. Most likely the doctor just obtained blood. It is as if no procedure had been done. The FNA has to be repeated (preferably by a more experienced aspirator). Please make sure that this is not confused with a benign diagnosis.